Samples—The Few, The Chosen

Code

Foundational Concepts

In this lecture, we're going to come to grips with the question of what only part of the data really means. So, this lecture is about the concept of a sample.

Hide

```
if (!caracas::has_sympy()) {
  caracas::install_sympy()
}
library(caracas)
library(sets)
library(tibble)
library(ggplot2)
library(dplyr)
```

Hide

```
def_sym("X", "Y")

cmStats <- modules::use("CmStats.R")$cmStats
```

With this lecture, we begin the investigation of how to infer features of the whole population from information about just some of the members of the population. A common scenario in which this plays out occurs when a poll is taken to estimate what proportion of voters favor which candidate.

Recall that the structure of our course is to view statistical analysis as having two basic parts: (1) how to describe, summarize, and organize a collection of data if we know all the data and (2) how to infer information about the whole population if we know only part of the data. The term population refers to the whole collection of people or things being considered. A sample is a subset of the total population, whether people, auto parts, or anything else we are investigating. We want to infer information about the whole population from information about the sample.

Voluntary response polls can lead to results that present a totally inaccurate view of the whole population. The simplest way to get a sample that is likely to be representative of the whole population is to just randomly pick a subset of the population. This is called a *simple random sample* (SRS).

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```
coins <- \() sample(c("T", "H"), size=2, replace = TRUE)

experiment = purrr::map(1:1000, \(s) coins());
```

Ask everyone to raise their hand if they either threw a head or they cheat.

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```
false_student <-  purrr::discard(experiment, \(s) s[1] == s[2])
true_student <-  purrr::keep(experiment, \(s) s[1] == s[2])
head_student <-  purrr::keep(experiment, \(s) s[1] == "H" &  s[2] == "H")
```

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```
counted <- list(false = length(false_student), true = length(true_student), head =
length(head_student))
```

• Suppose we do this experiment with 1000 students, and 800 raise their hands.

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```
cmStats$P(counted$head, counted$true)
```

```
[1] 0.4795918
```

We can estimate that about 60% of the students cheat because if 500 threw heads, then 300 of the 500 who threw tails cheated—about 60% (300/500).

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```
cmStats$P(300, 500)
```

```
[1] 0.6
```

In taking samples, our goal is to obtain samples from which we can accurately infer information about the whole population. That is, we want our samples to be representative of the whole population. Pitfalls to avoid in taking samples include avoiding biased samples and being wary of results obtained from voluntary response samples. We can obtain statistical information concerning sensitive information by using clever techniques in which no 43 individual needs to divulge sensitive secrets. The basic purpose of getting data from a sample is to infer information about the whole population. How that inference is made is the subject of the next lectures. # Concept and Applications

POINT ESTIMATES AND STANDARD ERROR

Descriptive statistics focus on summarizing characteristics about our data, such as calculating the sample mean or plotting histograms, and describe the dataset that's being analyzed but don't let us to draw any conclusions or make any interferences about the data. On the other hand, statistical inference uses our dataset to extract information about populations or answer real-world questions with certainty. It builds on the methods of descriptive statistics, but we can draw conclusions about the population based on data from a sample.

POINT ESTIMATES

Any time we have a random sample, we can calculate the sample mean, variance, and standard deviation. These values are called point estimates.

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```
head(Orange)
```

```
Grouped Data: circumference ~ age | Tree
  Tree  age circumference
1    1  118            30
2    1  484            58
3    1  664            87
4    1 1004           115
5    1 1231           120
6    1 1372           142
```

The dataset "Orange" has 35 rows and 3 columns of records of the growth of orange trees. Find a point estimate for the average orange tree growth and average tree circumference using the orange tree data.

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```
summary(Orange)
```

```
 Tree       age         circumference  
 3:7   Min.   : 118.0   Min.   : 30.0  
 1:7   1st Qu.: 484.0   1st Qu.: 65.5  
 5:7   Median :1004.0   Median :115.0  
 2:7   Mean   : 922.1   Mean   :115.9  
 4:7   3rd Qu.:1372.0   3rd Qu.:161.5  
       Max.   :1582.0   Max.   :214.0  
```

A point estimate for the average orange tree age would be 922.14 days, and a point estimate for orange circumference would be 115.85 millimeters.

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```
as_tibble_row(list(age =  mean(Orange$age), circ = mean(Orange$circumference)))
```

age <dbl>	circ <dbl>
922.1429	115.8571

1 row

The median is also a perfectly valid point estimate.

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```
as_tibble_row(list(age =  median(Orange$age), circ = median(Orange$circumference)))
```

age	circ
<dbl>	<dbl>
1004	115

1 row

Hide

```
NA
```

Notice how close the sample mean circumference is to the sample median of circumference: 115.85 compared to 115. And look how far off tree age estimates are from each other: 922.1 compared to 1004.

Which estimate is better? How do we decide between the 2?

STATISTICAL INFERENCE

Statistical inference is the process of making an estimate, prediction, or decision about a population based on a sample.

The rationale is that large populations make investigating every member impractical and expensive. It's easier and cheaper to take a sample and make estimates about the population from the sample. But those estimates aren't always going to be correct.

Statistical inference helps us do 2 main things: Draw conclusions about a population or process from sample data and provide a statement of how much confidence we can place in our conclusions.

ESTIMATION

The objective of estimation is to approximate the value of a population parameter on the basis of a sample statistic. We can estimate both points and intervals.

A point estimate of a parameter θ is a single number that can be regarded as a sensible value for θ. We find a point estimate by selecting a suitable statistic and computing its value from given sample data. The selected statistic is called the point estimator.

Point estimate is a value; point estimator is a statistic. Usually, we use to denote the point estimator of a parameter θ.

The "women" dataset in R contains the average heights and weights for American women aged 30 to 39. This is a sample of 15 women with those 2 variables: height in inches and weight in pounds.

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```
summary(women)
```

```
     height          weight     
 Min.   :58.0   Min.   :115.0  
 1st Qu.:61.5   1st Qu.:124.5  
 Median :65.0   Median :135.0  
 Mean   :65.0   Mean   :136.7  
 3rd Qu.:68.5   3rd Qu.:148.0  
 Max.   :72.0   Max.   :164.0  
```

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```
head(women)
```

	height <dbl>	weight <dbl>
1	58	115
2	59	117
3	60	120
4	61	123
5	62	126
6	63	129

6 rows

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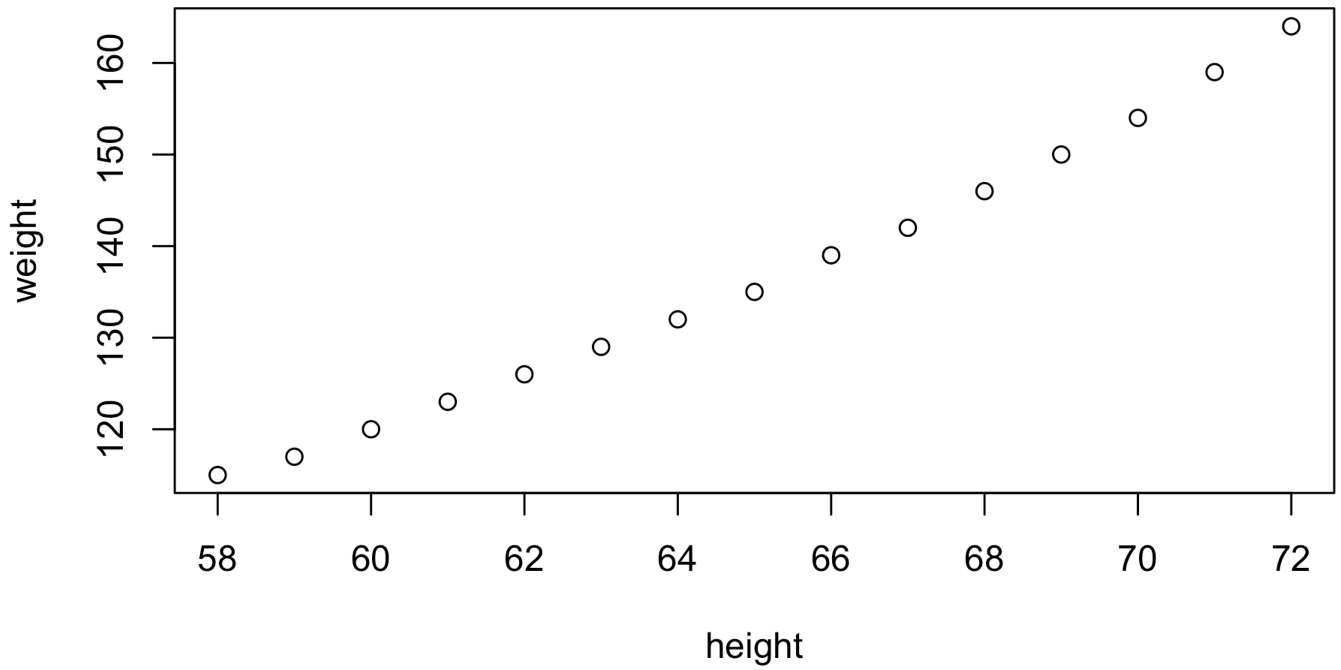
```
plot(women)
```



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```
list(mean = mean(women$weight), median = median(women$weight))
```

```
$mean
[1] 136.7333

$median
[1] 135
```

We can generate point estimates for the population parameters based on our sample statistics. What other estimators could we use to estimate the population mean (μ)?

Average of the extremes: . In fact, we're taking the range and halving it, so this gives us a measure of the center of the dataset.

Because we don't know how close we are to the true population parameter, we have to be strategic in selecting our point estimator. In other words, we want our point estimate to follow certain rules so that we know our sample estimate will be close to the population estimate.

CHOOSING AN OPTIMAL ESTIMATOR

How do we choose an optimal estimator?

A point estimator is a random variable, and its value varies from sample to sample. That variability is called the error of estimation.

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```
errro_of_estimation ~ phi:hat - phi
```

```
errro_of_estimation ~ phi:hat - phi
```

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```
# solving for phi_hat:
phi_hat ~ phi + error_of_estimation
```

```
phi_hat ~ phi + error_of_estimation
```

Which is the best one? We want to choose an estimator that is unbiased, one that is closest to the true value of the parameter θ on average, and that has minimum variance, or the one with the smallest estimation of error—the smallest variability, the least spread.

A point estimator is an unbiased estimator of the parameter θ if E(If is not unbiased, the difference E( ) – θ is called the bias of.

Other estimates exist, such as the range, or the average of the largest and smallest values. But these estimates are not unbiased and should not be used to estimate the mean.

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```
cmStats$mu_sym
```

```
c:    2        2
     i    i   n    n
   - -- + - + -- + -
     2    2   2    2
   -----------------
           N
```

Sample variance s^2 the unbiased estimator of the $n - 1$, population variance $\sigma 2$.

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```
sum(cmStats$X - cmStats$mu_sym)^2/ (cmStats$n - 1)
```

```
c:                           2
   /      2        2      \
   |     i     i   n     n|
   |   - -- + - + -- + -  |
   |     2     2   2     2|
   |X - --------------------|
   \           N           /
   ---------------------------
             n - 1
```

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```
cmStats$σ_2_sym
```

```
c:             2
   (-μ + sumₓ)
   ------------
        N
```

Suppose that X is a binomial with parameters n and p, with p unknown. We can show that the estimator = x / n is an unbiased estimator.

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```
E(X/n) ~ 1/n * E(x) ~1/n * np ~ p
```

```
E(X/n) ~ 1/n * E(x) ~ 1/n * np ~ p
```

For a normal distribution with mean μ and variance σ2, given a random sample of size n, X1, X2, … Xn, we can show that the sample mean is an unbiased estimator of μ.

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```
E(X_bar) ~ E(Sigma(X)/n) ~1/n * E(Sigma(X))
```

```
E(X_bar) ~ E(Sigma(X)/n) ~ 1/n * E(Sigma(X))
```

CONSISTENCY

An unbiased estimator is consistent if the variance of the estimator approaches 0 as our sample size (n) approaches infinity. In other words, the difference between the estimator and the population parameter becomes smaller as the sample size increases.

As another example, suppose instead that we estimate the mean by averaging the first and last values, X1 and Xn. This estimate is unbiased because

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```
E((X_1 + X_n)/2) ~E(X_1)/2 + E(X_n)/2 ~ mu/2 + mu/2 ~ mu
```

```
E((X_1 + X_n)/2) ~ E(X_1)/2 + E(X_n)/2 ~ mu/2 + mu/2 ~ mu
```

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```
as_func(cmStats$μ_hat_sym)
#(min(1:10) + max(1:10))/2
sum(cmStats$x)
``` ### still not finis!!!!!
```
```